
Assignment Title

Assignment XX

CourseID: CourseName

School, Semester

1. Question Components

A standard question:

[4.0 pts] What is $2 + 2$? Write a statement to evaluate in Python.

A multi-choice question:

[3.0 pts] What will the following Python expression be equivalent to?

$2 + 2$

- ☐ 4
- ☐ `len(np.array([1, 2, 3, 4]))`
- ☐ 2
- ☐ 6
- ☐ None of the above

[3.0 pts] You can display options with different column lengths.

- ☐ Option 1 ☐ Option 2 ☐ Option 3 ☐ Option 4

An answer bank:

A x^2	B A quadratic	C <code>x ** 2</code>
D A quartic	E $x \cdot x \cdot x$	F <code>pow(x, 3)</code>

[1.0 pt] What of the following functions are even?

- ☐ **A** ☐ **B** ☐ **C** ☐ **D** ☐ **E** ☐ **F** ☐ **G** ☐ **H** ☐ None of the above

A x^2	B Any linear function	C <code>x ** 2</code>
D Any quadratic function	E $x \cdot x \cdot x$	F <code>pow(x, 3)</code>
G $\sin(x)$	H Any quartic function	I $\cos(x)$

[1.0 pt] What of the following functions are even?

- ☐ **A** ☐ **B** ☐ **C** ☐ **D** ☐ **E** ☐ **F** ☐ **G** ☐ **H** ☐ **I** ☐ None of the above

2. [6.0 points] Second Section

Content in this second section does total up points

(a) [2.0 pts] Subquestion

(b) [2.0 pts] Subquestion

(c) [2.0 pts] Subquestion

3. Callouts

You may create a callout with an empty string "" to omit the title. Any non-special typed callout will be grey by default.

Special callout types (Definition, Formula, Method, Example)

Definition

A definition callout

Formula

A formula callout

Method

A method callout

Example

An example callout

4. Code Blanks

Complete the distance function below, which takes arrays of two predictor variables p1 and p2 and returns a distance between a new point row and each row in the training data.

```
def distance(p1, p2, row):  
    arr = np.array(row)  
    v1 = arr. _____ [A]  
    v2 = arr. _____ [B]  
    distances = _____ [C]  
    _____ [D]
```

(a) [2.0 pts] What function should be used in blanks [A] and [B] to retrieve the two items in the array?

(b) [2.0 pts] Fill in the blank [C], such that the distance function returns an **array** of Euclidean distances.

(c) [1.0 pt] Fill in the blank [D].